

DSA Class 2

→ Why do we need programming language?

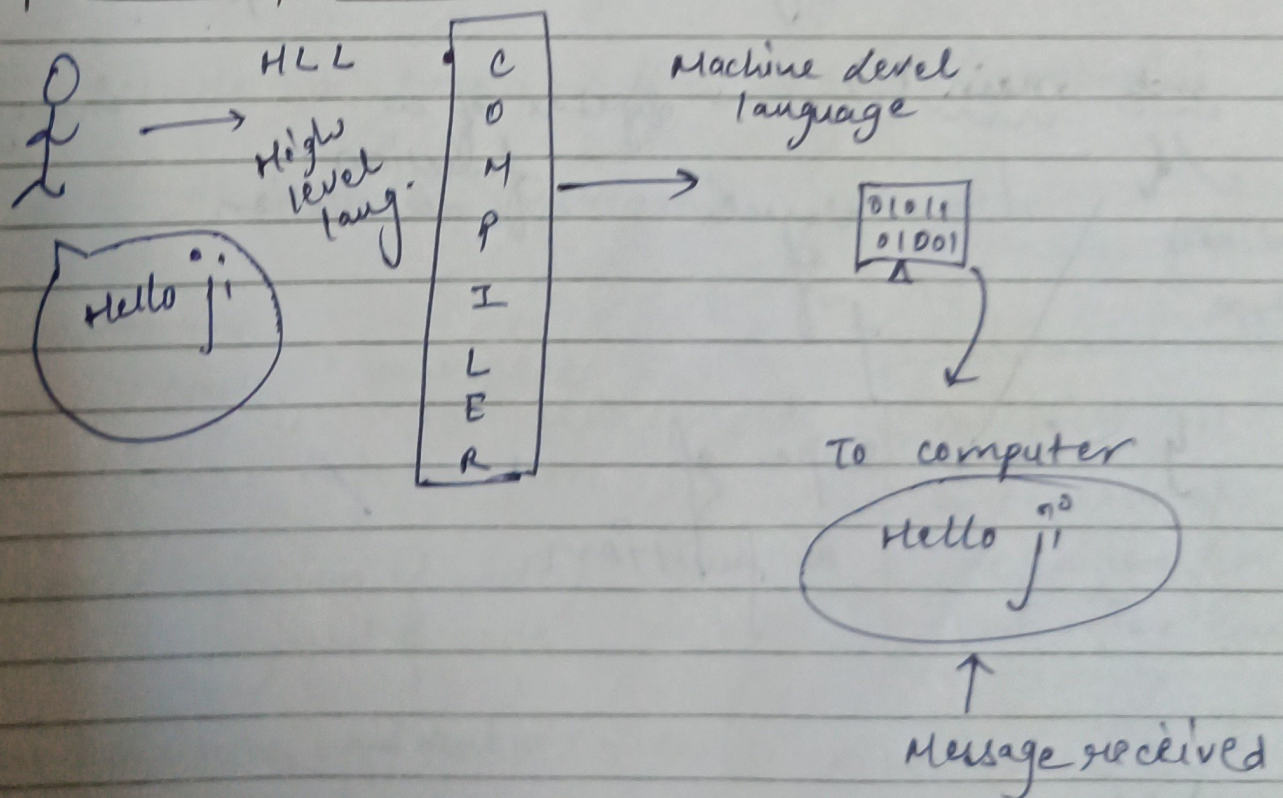
→ To get work done from computer, we need to communicate a computer to tell it, what to do or to instruct

→ To communicate a computer, we need a programming language

→ fixed set of rules for every programming language (Standardized syntax)

→ Every language has its own compiler/interpreter
translator tools.

Compilation process



⇒ Where to code

↘ Code Editor
or

IDE (Integrated Dev. Env)

The first c++ code :

Flowchart starts c

(start)

int main()
in code

code starts c

int main() } function

i/p

Function → An entity which is provided an ~~entity~~ block of code which one or more than one i/p is provides and maybe receive an o/p.

Enhance
reliability

int
return
type

main()

Syntax of a
function
declaration

int
integer

name
of
fn

scope of
a function

Q) Is int main() called by compiler?

A) Yes

why??

OOPS

can directly interact with memory
Speedful and efficient.

DATE

If I want to print something

Print $\rightarrow$ "charu"Use cout (character o/p)

i = 10

cout << i;

Used to display

cout << "charu" ;
 ↓ ↓ ↓
 to print or display syntax string end of line

We need to define cout using file "iostream"
 we need to include or import iostream

To include a header file use (iostream)

#include <iostream>

#include
 <header_file_name>

$\rightarrow$ std::cout namespace
 using namespace std;
 Additional information

namespace: portion of code/
 segment of code

where definition of identifier lies

compiler has
 a number of
 definition for
 identifier 'cout'

preprocessor directive

1) #include <iostream>
 2) using namespace std;
 int main() {

If we are not printing or making any i/p then all these two lines

→ Syntax
 cout << "Charu"; → String mandatory?
 identifier whose work is to print

y

Line 1: Before running code, include the mentioned header file

Line 2: Standard namespace will be used, where definition is mentioned.

int main() : Code will start from here.

code written in main function

To change line use [endl or "\n"]

cout << "Charu";
 cout << endl;
 cout << "Vicks";

OR

cout << "Charu";
 cout << "\n";
 cout << "Vicks";

OR

```
cout << "Charu" << endl;
```

```
cout << "Vicks";
```

To comment the code within main use

Ctrl + /

Comment exists on code but is not executed

→ int main () {

HW

return 0;

] Google its meaning

→ Successfully executed

```
#include <iostream>
using namespace std;
```

```
int main ()
```

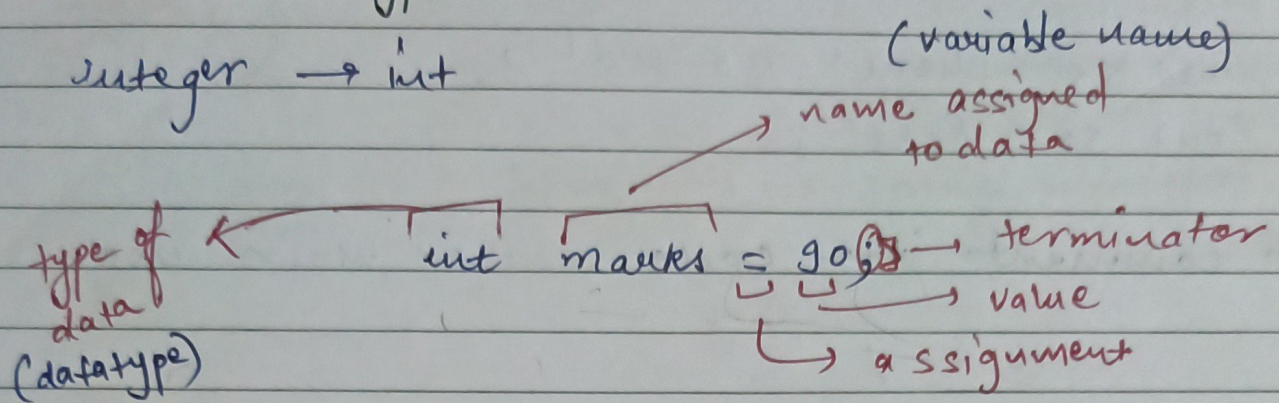
```
{
```

```
    cout << "Charu";
    return 0;
```

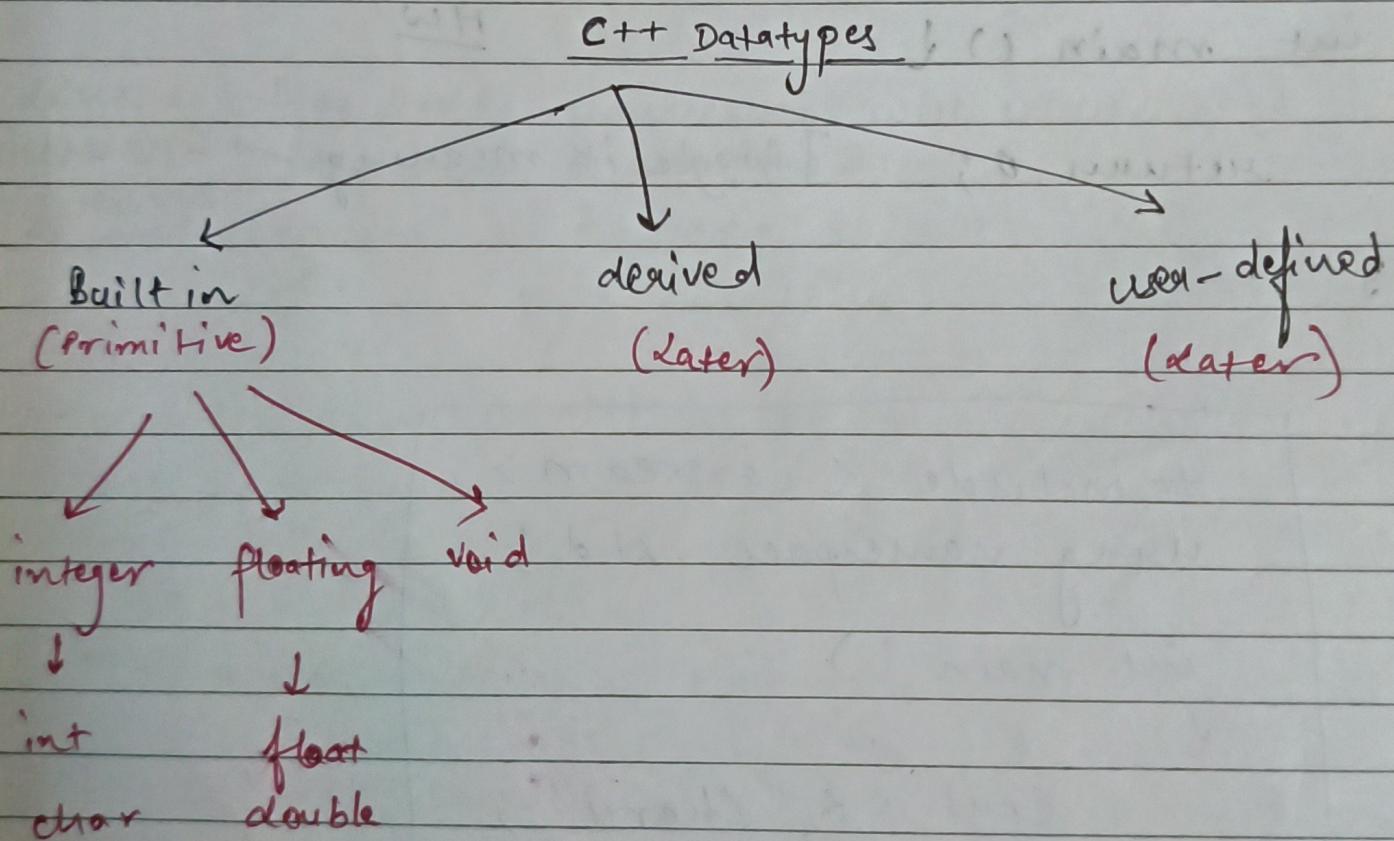
```
}
```

return 0 means program executed successfully

Variables and Datatypes.



Data can be of diff. types.



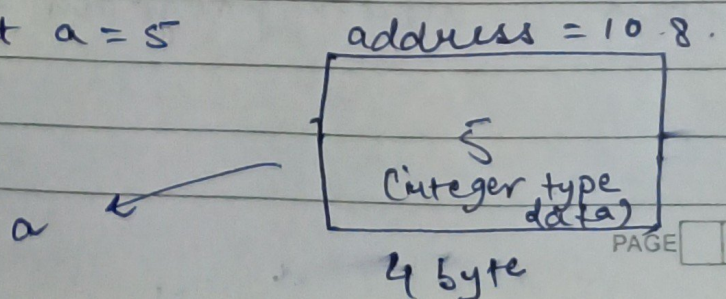
Inside memory:

1 byte = 8 bits.

Bits $\rightarrow$ 0/1

int $\rightarrow$ 4 byte (size)

$\Rightarrow$ int a = 5



void (empty)

void main() {

} (nothing will be returned from here)

What is a variable?

- Name given to memory location
- Value stored in variable can be changed
- Type of data stored in variable is determined by datatype

- ↳ Tells the type of data
- ↳ Space occupied by that data.

Size occupied by different datatypes: ⇒

Declaration ⇒ int num; (only it is informed that an integer type block is reserved)

Initialisation ⇒ int num = 50; (Block is defined as well as a value is given)

Size occupied by integer is 4 bytes but maybe it may contain 2 bytes in some machines (Machine dependent)

To check size of datatype: sizeof (variable name)

int main() {

int num = 50;

cout << num << endl;

cout << sizeof(num) << endl;

O/P:

50

4

Datatype	(Bytes) Size (64 bit)	(Bytes) Size (32 bit)
int	4	4
char	1	1
float	4	4
long	8	4
short	2	2
long long	8	8
double	8	8

→ Range will be ~~discussed~~ understood later

when we try to print value of a declared variable which hasn't been assigned a value, it prints some garbage value.

Variable naming conventions

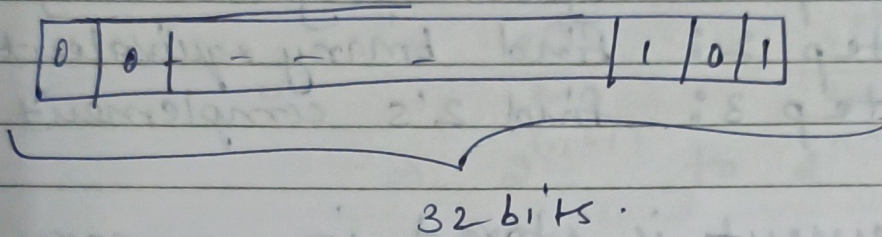
- Variable name should be in such a way that it ~~do~~ gives a meaning
- Begins with alphabet
- No spaces in variable name.
- Digits can be used but not in beginning.
- No special symbol, only underscore () allowed
- Case sensitive i.e. A is different from a
- No keyword or command should be used as a variable name.

How data is stored?

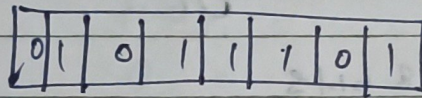
int $\rightarrow$ 4 byte $\rightarrow$ 32 bits

eg) int a = 5; $\rightarrow$ Binary value of 5 will be stored in 32 bits

Integer



character



Every character is mapped with a integer value

ASCII value

while storing character, value of that is turned into binary and that is stored in 8 bits
(Binary equivalent of)

Boolean

Either stores 0/1

But still it takes 1 byte i.e. 8 bits.
It wastes 7 bits

But why?

Smallest addressable space is of 1 byte in machine

For negative numbers:

↳ 1's complement repⁿ

↳ 2's complement repⁿ

$\Rightarrow$ int a = -5

↳ Step 1: Ignore Negative Sign

↳ Step 2: Find binary equivalent

↳ Step 3: find 2's complement

This 2's complement is stored in memory

Ranges of different data types:

1 byte $\rightarrow$ 8 bits

for char, $\Rightarrow$ ~~8 bits~~ Total 2^8 characters can be created

Range: 0 to $2^8 - 1$

~~8 bits~~ Unsigned

Generic formula for n bits $\Rightarrow$ Total combinations: 2^n
Range: 0 to $2^n - 1$

for integer, Total combⁿ: 2^{32}

(4 bytes $\rightarrow$ 32 bits) Range: 0 to $2^{32} - 1$

for long $\rightarrow$ 8 byte $\rightarrow$ 64 bits

Total combⁿ : 2^{64}

Range : 0 to $2^{64} - 1$

~~Unsigned~~ Signed (n bits)

Total combinations : 2^n

Range : $\frac{2^n}{2}$ to $\frac{2^n}{2} - 1$

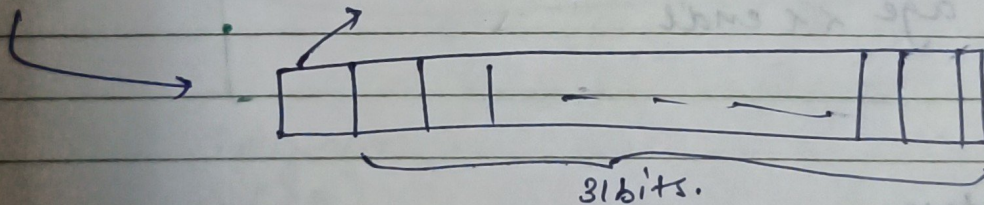
$\Rightarrow 2^{(n-1)}$ to $2^{(n-1)} - 1$

Signed vs Unsigned Data :

$\downarrow$
+ve
-ve
0

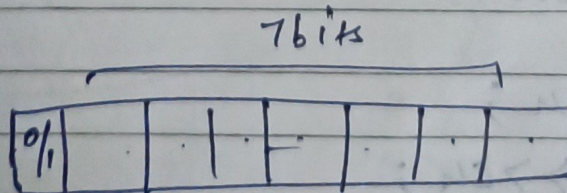
$\downarrow$
+ve
0

MSB | leftmost bit (sign bit)



If MSB = 0 $\Rightarrow$ Positive
MSB = 1 $\Rightarrow$ Negative

for visualising



Range : for +ve $\rightarrow 2^7$ (0 $\rightarrow$ 127)

for -ve $\rightarrow 2^7$ (-128 $\rightarrow$ -1)

Operators

- 1) Arithmetic : $+, -, *, / , \cdot$
For arithmetic operation
- 2) Relational : $>, <, <=, >=, !=, ==$
Used for comparisons
- 3) Assignment : $=$
Used for assigning value
- 4) Logical : $\&\&, \|\|, !$
Used when multiple conditions are there

! `age = 0;`
 `cout << ! age << endl;`

$\Rightarrow$ o/p : 1

`age = (any number) 10;` $\rightarrow$ o/p : 0
`cout << ! age << endl`

$\|\|$: Any one true
 $\&\&$: Both True

Input Use cin.

1) eg ~~char~~ `ch;`
 `cin >> ch;`
 `cout << " ch : " << ch << endl;`

o/p : K
 ch : K

NOTES

DATE

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```
2) int marks;  
cout << "Enter marks here: " << endl;  
cin >> marks;  
cout << "Marks: " << marks << endl;
```

Read

```
cin >> variable_name;
```

Print

```
cout << "string" << endl;
```